Accenture Australia Pty Ltd

Spatial WA – Next Generation Spatial Cadastre (NGSC)

End of project report

Prepared by: Surround Australia Pty Ltd (SURROUND)

Date: 2026-11-02

Table of contents

1 1. Executive summary 1

2 2. Project context, scope, and approach 1

2.1 2.1 NGSC programme context and objectives 1

2.2 2.2 Contractual scope and responsibilities 2

2.3 2.3 Requirements analysis methodology 2

2.4 2.4 Machine-readable specifications and technical outputs 3

2.5 2.5 Engagement, collaboration, and knowledge transfer 4

3 3. Deliverables and outcomes 4

3.1 3.1 Summary of the five deliverables 6

3.2 3.2 Coverage of the 34 NGSC requirements 6

3.3 3.3 Technical and architectural contributions 6

3.4 3.4 Acceptance criteria and validation readiness 6

3.5 3.5 Status of outputs and remaining dependencies 6

4 4. Consolidated technical and strategic findings 6

4.1 4.1 3D CSDM alignment and WA profile development 6

4.2 4.2 Cadastral data exchange and interoperability 6

4.3 4.3 3D geometry, topology, and boundary representation 6

4.4 4.4 Metadata, data quality and standards compliance 6

4.5 4.5 Cadastral workflows, APIs, and system integration 6

4.6 4.6 Cross-cutting dependencies and architectural decisions 6

5 5. Strategic assessment: strengths, gaps, opportunities, and risks 6

5.1 5.1 Strengths and capabilities established 6

5.2 5.2 Capability gaps and areas requiring further development 6

5.3 5.3 Opportunities for further value creation 6

5.4 5.4 Risks to delivery and long-term sustainability 6

6 6. Opportunities to strengthen NGSC delivery and capability 6

6.1 6.1 Prioritisation principles 7

6.2 6.2 WA 3D CSDM profile and standards development 7

6.3 6.3 Validation, testing, and quality assurance 7

6.4 6.4 Data transformation and migration 7

6.5 6.5 3D cadastre implementation and demonstrators 7

6.6 6.6 Integration, governance, and capability development 7

7 7. Implementation roadmap and engagement opportunities 7

7.1 7.1 Immediate priorities and decision points 7

7.2 7.2 Medium-term development opportunities 7

7.3 7.3 Longer-term strategic opportunities 7

7.4 7.4 Potential delivery and collaboration arrangements 7

8 8. Conclusions and recommended next steps 7

8.1 8.1 Overall assessment 7

8.2 8.2 Decisions and actions recommended **7**

8.3 8.3 Pathway to future collaboration **7**

9 References **8**

Acronyms and Terms

Acronym	Description
AI	Artificial Intelligence; in this document, automated methods (including ML/LLMs) used to support planning, reasoning, and decision support.
AIS	Automatic Identification System; vessel tracking broadcasts used here as an input dataset for collision-risk and CPA computations.
AOI	Area of Interest; The area that a Commander/Tasker is concerned about because what happens in the area could affect the task.
API	Application Programming Interface; a software interface used for system-to-system communication.
AWS	Amazon Web Services; referenced as the hosting environment (e.g., AWS API Gateway) for prototype services.
AUVRP	Requirement identifier prefix used in this report (e.g., AUVRP-01) to label and track prototype demonstration components.
AZ	Availability Zone; an isolated, physically separated data centre within an AWS region.
CMEMS	Copernicus Marine Environment Monitoring Service; referenced as the source of global ocean physics analysis/forecast ocean current products.
CPA	Closest Point of Approach; a collision-assessment concept for estimating the minimum separation between two moving objects and when it occurs.
CRS	Coordinate Reference System; defines how coordinates map to locations on Earth (and how metrics such as distance or direction are computed).
DAG	Directed Acyclic Graph; used to describe a workflow graph (e.g., in Airflow) with ordered tasks.
DCPA	Distance to Closest Point of Approach; the minimum predicted distance between two moving objects during their encounter.
DevOps	DevOps information refers to the practice of combining software development (Dev) with operations and maintenance (Ops).
Docker	A standalone executable software package; Used to ensure a process runs consistently across any computing environment.
EDR	Environmental Data Retrieval; an OGC API family for querying/serving environmental/coverage data.
EC2	Amazon Elastic Compute Cloud; a core AWS service used to host the prototype services.
EARS	Easy Approach to Requirements Syntax; a structured pattern set for writing clear, testable requirement statements (e.g., ubiquitous, state-driven, event-driven).
EPSG	European Petroleum Survey Group; registry of geodetic/CRS identifiers (EPSG codes) such as EPSG:4326 (WGS84) and EPSG:9473 (GDA 2020 Australian Albers).
GDB	Geodatabase; commonly an ESRI file geodatabase format used to store GIS datasets.
GeoJSON	A JSON-based geospatial data interchange format (features + geometries), referenced for returning routes as a FeatureCollection.
GeoTIFF	Geographic Tagged Image File Format; a TIFF raster image with embedded georeferencing metadata.
HA	High Availability; designing applications to ensure high uptime through redundancy and clustering; not supported in prototype but enabled by component model.

Acronym	Description
HTTPS	Hypertext Transfer Protocol Secure; HTTP over TLS for encrypted network communication.
ICT	Information and Communications Technology; referenced in the context of Defence ICT networks and security standards.
JPG	JPEG image format; referenced as a supported data type for ingestion/fusion.
KG	Knowledge Graph; a graph-based representation of entities and relationships used here for flexible information capture and provenance.
KML	Keyhole Markup Language; an XML-based geospatial format commonly used for maps and features.
LLM	Large Language Model; an AI model class used here for assisted interpretation, reasoning, and explanation in mission planning.
m/s	Metres per second; metric unit of speed referenced for consistent calculation/output.
MP4	MPEG-4 Part 14; a digital multimedia container format (video/audio).
NetCDF	Network Common Data Form; a scientific data format commonly used for gridded environmental/ oceanographic datasets.
NBI	Northbound Interface; a lower-level network component that communicates upwards with higher-level applications often using REST APIs protocols. In simple terms, it's a connection that lets an application talk to the system's brain to obtain computational power and information.
OFFICIAL	A protective marking / classification level used in Australian Government information handling (referenced here as OFFICIAL-level data).
OGC	Open Geospatial Consortium; global standards organization producing geospatial interoperability standards including OGC APIs.
OPTASK	Operational Tasking (document); referenced as an input document type used alongside signals for mission/task interpretation.
PostGIS	Spatial extension for PostgreSQL adding geospatial types, functions, and spatial indexing for network and route management.
PROV	W3C PROV; the W3C provenance data model used to record traceable inputs, processes, and outputs.
RAN	Royal Australian Navy.
RBAC	Role-Based Access Control; access permissions are assigned based on user roles (e.g., planner, operator, administrator).
REST	Representational State Transfer; architectural style for web services and APIs used for component integration.
RFI	Request for Information; a formal request used to gather mission-relevant information and inputs.
SaaS	Software as a Service; a cloud computing model where software applications are delivered and accessed over the internet on a subscription basis, rather than being installed locally.
SME	Subject-Matter Expert (also appears as SMEs); domain experts supporting definition, review, validation, and assessment.
SPARQL	SPARQL Protocol and RDF Query Language; a query language for RDF/graph data, used here to query the Knowledge Graph.
SRS	System Requirements Specification; the document defining intended capabilities, constraints, and testable requirements.

Acronym	Description
STANAG	Standardisation Agreement (NATO); referenced as a formatted signal type that may include structured and free-text elements.
TCPA	Time to Closest Point of Approach; the predicted time until the moving objects reach their minimum separation.
TLS	Transport Layer Security; cryptographic protocol used to secure communications (e.g., HTTPS/TLS).
u	Eastwards vector component of ocean current velocity in the referenced NetCDF current products.
UI	User Interface; the user-facing interface, constrained here to be map-centric.
UI/API reachability	The UI and API can be accessed from the intended network and return expected responses (page loads for the UI; endpoint responds for the API).
UUV	Uncrewed Underwater Vehicle; the platform type whose missions/routes are being planned.
UUVRP	Uncrewed Underwater Vehicle Route Planning; the prototype/trial capability described in the report.
UTC	Coordinated Universal Time; the time standard used for timestamps in examples and workflow parameters.
UX/UI	User Experience / User Interface; design concerns around usability and interaction patterns for the system.
v	Northward vector component of ocean current velocity in the referenced NetCDF current products.
VM	Virtual Machine; a software environment that runs on top of a host operating system (e.g., AWS EC2). Similar to Docker, but runs on top of the host OS rather than a containerised environment.
VPC	Virtual Private Cloud; is a logically isolated, virtual network within the Amazon Web Services (AWS) cloud.
W3C	World Wide Web Consortium; produces web standards including provenance standards (e.g., PROV).
WGS84	World Geodetic System 1984; a global geodetic datum commonly associated with geographic coordinates (lat/lon).
XTF	eXtended Triton Format; commonly used as a side-scan sonar data format (listed here as an ingestible hydrographic format).
XYZ	A simple coordinate listing format (often ASCII) representing points as X, Y (and optionally Z) values; listed as a supported data type.

Revision History

Version	Status	Date	Author	Description
0.1	Draft	2026-10-15	Andrew Hunter	Initial TOC

This document is complete as of the date of presentation to RAN in satisfaction of the Milestone #6 deliverables.

1. Executive summary

2. Project context, scope, and approach

This section outlines the context in which SURROUND was engaged to support the Western Australian Next Generation Spatial Cadastre (NGSC) project, the scope of its responsibilities, and the approach adopted in delivering the contracted work.

It describes SURROUND's relationship with Accenture and Landgate, the methodology applied across the five project deliverables, and the distinction between cadastral information modelling and NGSC system implementation responsibilities. It also summarises the collaborative approach to requirements analysis, technical specification development, and knowledge transfer.

Together, these provide the context for assessing the work undertaken, the technical outputs produced, and the opportunities for further development identified through the engagement.

2.1 NGSC programme context and objectives

The Western Australian Next Generation Spatial Cadastre (NGSC) project forms part of Landgate's broader Spatial WA programme. The project seeks to modernise the management of cadastral information, including the transition from predominantly two-dimensional cadastral representations to a spatial cadastre capable of supporting three-dimensional parcels, improved data quality, automated validation, and integration with contemporary spatial information systems.

Accenture is the prime contractor responsible for delivery of the NGSC solution to Landgate. Accenture engaged SURROUND Australia to provide specialist cadastral information modelling, standards development, and technical advisory services, with particular emphasis on the Intergovernmental Committee on Surveying and Mapping (ICSM) 3D Cadastral Survey Data Model (3D CSDM).

SURROUND's primary objective was to assist Accenture and Landgate in translating the NGSC business and functional requirements into well-defined, standards-aligned information requirements that could be implemented, tested, and maintained. This included identifying where existing national and international standards could be applied, where Western Australian requirements warranted extensions or additional specification, and where the proposed functionality depended on decisions concerning NGSC system design or business processes.

An important consideration throughout the engagement was that the 3D CSDM is an information exchange standard, rather than a specification for the complete NGSC system. The work therefore sought to distinguish the responsibilities of the cadastral information model from those of the applications, services, and business processes required to implement and maintain the spatial cadastre.

The intended outcome was not just to document individual requirements, but to establish a more consistent and reusable standards-based foundation for NGSC development, with opportunities for adoption beyond the immediate Western Australian implementation.

2.2 Contractual scope and responsibilities

SURROUND was engaged under a 12-month subcontract to Accenture, within the broader NGSC delivery programme. The contracted services covered specialist data design, requirements analysis, co-design, development of CSDM profiles and supporting specifications, identification of technical constraints, validation, and definition of potential acceptance criteria.

The Statement of Work distinguished between requirements for which SURROUND had responsibility for leading specified CSD and 3D CSDM-related outputs and requirements for which SURROUND provided specialist support to the Accenture implementation team. This distinction is important when assessing the work undertaken and the associated delivery responsibilities.

For the nominated CSD and 3D CSDM requirements, the contract identified technical outputs including machine-readable schemas, ontologies, example datasets, validation rules, and transformation specifications, where applicable. It also required documentation, test cases, and acceptance criteria to support subsequent solution development and verification.

For supporting requirements, SURROUND's role was primarily to assist in interpreting cadastral and 3D requirements, advise on information modelling and relevant standards, identify technical limitations and dependencies, and contribute to the definition of appropriate acceptance criteria.

Accenture retained overall responsibility for programme management, solution architecture, system development, integration, implementation, and delivery to Landgate. Landgate, as the client and business owner, retained responsibility for confirming business requirements, jurisdictional policy, and operational decisions. SURROUND's technical advice and specifications were intended to inform these activities rather than replace the detailed design and implementation responsibilities of the NGSC delivery team.

This allocation of responsibilities was particularly relevant where a requirement involved both information modelling and operational functionality. For example, the 3D CSDM can define and validate cadastral information exchanged with the NGSC, but the implementation of user interfaces, workflows, application programming interfaces, database services, and operational controls remains a responsibility of the NGSC solution.

The distinction also informed how unresolved technical and business matters were treated. Where a requirement depended on an agreed workflow, system architecture, data ownership arrangement, or jurisdictional rule, the dependency was identified rather than assuming a design decision on behalf of Accenture or Landgate.

2.3 Requirements analysis methodology

The work was undertaken through five deliverables, aligned with the staged NGSC programme requirements. Together, these addressed 34 requirements spanning cadastral data growth, data standards, 3D spatial representation, survey information, boundary management, metadata, validation, system interfaces, and associated business functions.

The methodology evolved during the engagement but retained a consistent approach to interpreting requirements and identifying the contribution of the 3D CSDM. For each requirement, the analysis

generally addressed six related considerations: 1. Requirement interpretation: Establish the intended business capability, distinguish functional outcomes from proposed implementation approaches, and identify ambiguities or gaps in the requirement. 2. NGSC responsibilities: Identify the functionality expected of the NGSC solution, including relevant applications, services, workflows, and operational controls. 3. 3D CSDM alignment: Assess whether existing data models and standards support the requirement and identify any necessary extensions, information structures, or implementation patterns. 4. Technical constraints: Consider data availability, geometric and topological integrity, compatibility with existing survey information, validation, transformation, and interoperability. 5. Potential acceptance criteria: Propose testable outcomes and verification approaches, without unnecessarily prescribing implementation technology or detailed solution design. 6. Dependencies and decisions: Record outstanding policy, business, architectural, and technical matters requiring further consideration by Accenture, Landgate, or the NGSC implementation team. The earlier deliverables included more detailed examination of individual data standards and technical requirements, while later deliverables adopted a consolidated reporting structure incorporating the original WA requirement, analysis, NGSC System Perspective, 3D CSDM Perspective, and potential acceptance criteria.

A deliberate feature of this approach was the separation of the NGSC System Perspective from the 3D CSDM Perspective. SURROUND prepared the requirement analysis and cadastral information modelling perspective, with the NGSC System Perspective reserved for completion by the implementation team. This provided a basis for coordinated design while avoiding assumptions about implementation decisions that had not yet been made.

The analyses also recognised that not all requirements could be fully resolved through information modelling. Matters such as metadata governance, lodgement and approval workflows, system ownership, historical survey information, and the precise interpretation of certain 3D geometric operations required additional business or technical decisions. Identifying these dependencies was considered an important outcome of the analysis, as it provided greater clarity regarding subsequent design and implementation work.

2.4 Machine-readable specifications and technical outputs

In parallel with the requirements analysis, development work was undertaken to support machine-readable specifications and to investigate practical implementation approaches.

This included work on the Western Australian 3D CSDM profile and supporting common model components; development and refinement of controlled vocabularies; structured cadastral examples; topology and validation functions; and prototype capabilities for visualisation, reporting, and survey data processing.

The work was managed through GitHub repositories, allowing specifications, example datasets, validation logic, and supporting software to be developed and tested in a repeatable manner. These artefacts provide an implementation-oriented complement to the written requirements analyses and help demonstrate how the proposed information structures and constraints can be applied.

The principal technical workstreams are represented by the WA 3D CSDM profile, ICSM vocabularies, common CSDM components, Topo Feature model, and topology validation functions. Supporting (proof of concept) developments included horizontal section and 3D viewers, reporting components, and survey

calculation utilities. These are discussed further in later sections of this report, together with their applicability to individual NGSC requirements.

2.5 Engagement, collaboration, and knowledge transfer

SURROUND's engagement was structured to support Accenture's delivery responsibilities through specialist advice, collaborative requirements interpretation, and development of standards-based technical outputs. The intended approach involved co-design with Landgate and Accenture and its spatial implementation team to clarify business requirements, policy considerations, and acceptance expectations.

Collaboration was important because cadastral requirements frequently span multiple areas of responsibility. Survey and land administration practices define the meaning of the information, standards establish how it can be represented and exchanged, and the NGSC implementation determines how that information is managed and used operationally.

The requirements reports and associated technical artefacts were developed to provide a common reference for these discussions. Proposed acceptance criteria, documented dependencies, and practical examples were intended to support later design workshops, implementation planning, and verification.

Knowledge transfer was also supported through documented specifications, shared development repositories and reusable examples and validation components. The technical work adopted open standards and the OGC Building Blocks approach, allowing specifications and their supporting examples to be published, tested, and maintained independently of any particular NGSC application.

This approach sought to provide Accenture and Landgate with more than individual requirement interpretations. It established a body of reusable cadastral information modelling and validation work that can support continued NGSC development, future refinement of the Australian 3D CSDM, and wider adoption of consistent cadastral information standards.

The distinction between completed analysis, supporting technical development and outstanding implementation decisions remains important. The outputs of this engagement provide a foundation for subsequent design and implementation, but their value will ultimately depend on their adoption, integration, and verification within the NGSC solution.

3. Deliverables and outcomes

Deliverable	Requirements	Principal focus
1	1	NGSC data growth, scalability and 3D cadastral capability
2	10	Geographical naming, ABS Data Quality Framework and government standards
3	15	Cadastral editing, 3D visualisation, data exchange, workflows and APIs
4	6	Survey control, metadata, plan lodgement, validation and boundary relationships
5	2	Metadata feedback workflows and 3D boundary alignment

Deliverable	Requirements	Principal focus
Total	**34**	

3.1 Summary of the five deliverables

3.2 Coverage of the 34 NGSC requirements

3.3 Technical and architectural contributions

3.4 Acceptance criteria and validation readiness

3.5 Status of outputs and remaining dependencies

4. Consolidated technical and strategic findings

4.1 3D CSDM alignment and WA profile development

4.2 Cadastral data exchange and interoperability

4.3 3D geometry, topology, and boundary representation

4.4 Metadata, data quality and standards compliance

4.5 Cadastral workflows, APIs, and system integration

4.6 Cross-cutting dependencies and architectural decisions

5. Strategic assessment: strengths, gaps, opportunities, and risks

5.1 Strengths and capabilities established

5.2 Capability gaps and areas requiring further development

5.3 Opportunities for further value creation

5.4 Risks to delivery and long-term sustainability

6. Opportunities to strengthen NGSC delivery and capability

6.1 Prioritisation principles

6.2 WA 3D CSDM profile and standards development

6.3 Validation, testing, and quality assurance

6.4 Data transformation and migration

6.5 3D cadastre implementation and demonstrators

6.6 Integration, governance, and capability development

7. Implementation roadmap and engagement opportunities

7.1 Immediate priorities and decision points

7.2 Medium-term development opportunities

7.3 Longer-term strategic opportunities

7.4 Potential delivery and collaboration arrangements

8. Conclusions and recommended next steps

8.1 Overall assessment

8.2 Decisions and actions recommended

8.3 Pathway to future collaboration

References